



Practical-7 : Spark Application word count

```
import os
import subprocess
import time
```

```
os.environ["JAVA_HOME"] = r"C:\Program file\Java\jdk-11"
```

```
os.environ["SPARK_HOME"] = r"C:\spark\spark-3.5.1-bin-hadoop3"
```

```
os.environ["HADOOP_HOME"] = r"C:\hadoop"
```

```
os.environ["PATH"] += ";" + os.environ["SPARK_HOME"] + r"\bin"
```

```
os.environ["PATH"] += ";" + os.environ["HADOOP_HOME"] + r"\bin"
```

```
subprocess.Popen([os.environ["SPARK_HOME"] + r"\bin\start-master.cmd"], shell=True)
```

```
time.sleep(5)
```

```
scala_code = """
```

```
import org.apache.spark.sql.sparkSession
Object wordCount
```

```
{
```

```
def main(args: Array[String]):
```

```
Unit = {
```

```
val spark = sparkSession.builder
```



```
• val appName ("wordCountExample")
• master ("spark://localhost:7077")
• getOrCreate()
```

```
val sc = spark.sparkContext
val textfile = sc.textFile ("sample.txt")
val counts = textfile.flatMap (line =>
    line.split (" "))
    .map (word => (word, 1))
    .reduceByKey (_+_ )
counts.collect().foreach (println)
spark.stop()
```

```
with Open ("wordCount.scala", "w") as f:
    f.write (scala_code)
```

```
subprocess.run ["scalac", "-classpath", os.environ["SPARK_HOME"] + "/lib/*", "wordCount.scala",
    check = True]
```

```
subprocess.run ["
```

```
    "spark-submit",
    "--class", "wordCount",
    "--master", "spark://localhost:7077",
    "wordCount.jar"]
```